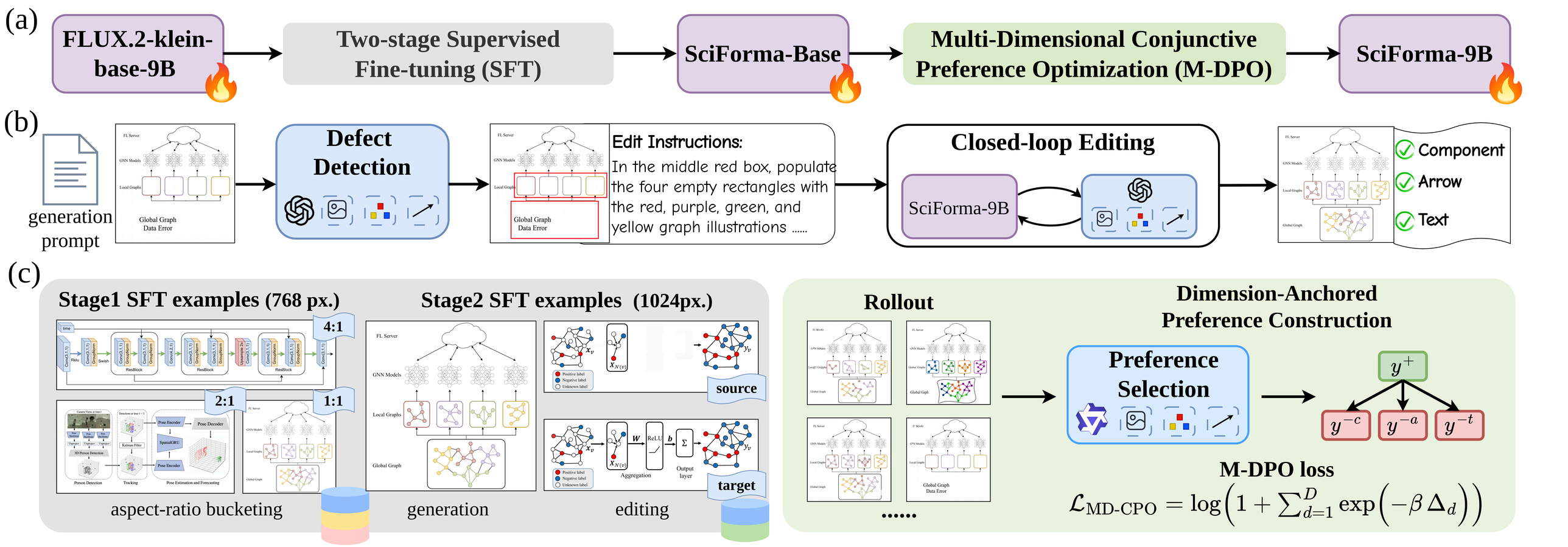
(a)
FLUX.2-klein-base-9B
Two-stage Supervised Fine-tuning (SFT)
SciForma-Base
Multi-Dimensional Conjunctive Preference Optimization (M-DPO)
SciForma-9B
(b)
generation prompt
FL Server
GNN Models
Local Graphs
Global Graph
Data Error
Defect Detection
Edit Instructions:
In the middle red box, populate the four empty rectangles with the red, purple, green, and yellow graph illustrations .....
Closed-loop Editing
SciForma-9B
Component
Arrow
Text
(c)
Stage1 SFT examples (768 px.)
4:1
2:1
1:1
aspect-ratio bucketing
Stage2 SFT examples (1024px.)
generation
source
target
editing
Positive label
Negative label
Unknown label
Aggregation
Output layer
Rollout
.....
Dimension-Anchored Preference Construction
Preference Selection
M-DPO loss
$\mathcal{L}_{\text{MD-CPO}} = \log\left(1 + \sum_{d=1}^{D} \exp\left(-\beta\,\Delta_d\right)\right)$